

Non-singular predefined time sliding mode control for quadrotor UAV with actuator fault

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Abstract

Aiming at the problems of unknown disturbances and actuator failures during the flight of quadrotor unmanned aerial vehicle (UAV), a non-singular predefined time sliding mode controller is designed for quadrotor UAV with actuator failures. Firstly, the actuator failure model is analyzed, and the quadrotor UAV with failure model is constructed. Secondly, the uncertain terms, external disturbances, and actuator failures are regarded as lumped disturbances, and the predefined time disturbance observer is designed to estimate the lumped disturbances. The designed observer can ensure the tracking of the lumped disturbances within the predefined time. Finally, a non-singular predefined time segment sliding surface and an improved segmented exponential convergence law are designed. The proposed sliding surface avoids the singular problem caused by the traditional predefined time sliding surface, while the segmented exponential convergence law suppresses the chattering of the control input. The stability of the designed observer and controller was proved through Lyapunov theory. The simulation experiments show that under the proposed control method, the tracking performance of the quadrotor UAV can complete convergence within the predefined time. Meanwhile, during the actuator failure period, the proposed control method outperforms the traditional predefined time sliding mode control and non-singular terminal sliding mode control in terms of robustness.

Keywords

quadrotor UAV, actuator failures, predefined time, disturbance observer, robustness

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Introduction

Quadrotor UAV has vertical take-off and landing, good stability and simple structure, which enables it to be widely used in military industry, agriculture and forestry, and various civilian fields.¹ However, due to the underdrive and strong coupling characteristics of quadrotor UAV, achieving the efficient flight control of quadrotor UAV is challenging.² Therefore, it has attracted many scholars to devote themselves to the control research of quadrotor UAV and proposed many effective control methods. Common linear control algorithms include PID control algorithm^{3–5} and LQR control algorithm,^{6,7} among which PID algorithm is the most widely used in the field of quadrotor control. Due to the increased complexity of the control target and the influence of external interference, linear control algorithms may encounter problems such as insufficient control accuracy and stability. Therefore, nonlinear control methods such as adaptive control^{8–11} and sliding mode control^{12–15} have been widely studied. These algorithms have certain adaptability to the

uncertainty of the model and unknown disturbances, improving the robustness of the quadrotor UAV control system.

With the increasing requirements for the control performance of quadrotor, it is expected that quadrotor has a faster response. Firstly, the finite-time control algorithm was carried out,^{16–19} which can achieve finite-time convergence to the desired value. Subsequently, the fixed-time control algorithm was developed,^{20,21} whose advantage is that the boundary of the stable time is a constant and independent of the initial conditions. In Hassani et al.,²² a super-twisted nonsingular terminal sliding mode controller was proposed, which ensures the completion of trajectory tracking within a finite time. Meanwhile, a nonlinear

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disturbance observer was designed to estimate unknown disturbances, enhancing the robustness of the system. In Shao et al.,²³ a fixed-time extended state observer was proposed using two kinds of polynomial feedback to estimate the velocity state quantity and uncertainty terms, thereby designing a fixed-time controller without velocity to reduce the influence of chattering.

In recent years, predefined time control algorithms have been designed, which advantage is that the boundary of convergence time can be set through parameters. The advantage has promoted the application of predefined time control in engineering. In Cui et al.,²¹ a predefined time observer was designed to estimate external disturbances by using the predefined time control algorithm, and a predefined time sliding mode controller was designed to control the spacecraft attitude, which effectively improved the convergence. In Labbadi et al.,²⁴ a predefined time sliding mode controller was developed to achieve the position and attitude control of the quadcopter, which improve the convergence speed of the quadrotor UAV, but there was a singularity problem in the designed sliding mode surface. In Qin et al.,²⁵ a predefined time sliding mode control method was proposed. To suppress the buffeting of predefined time sliding mode control, a variable gain hyper warp algorithm with additional linear terms was designed as a switch item.

The research on the above-mentioned control scheme has significantly improved the tracking performance of quadrotor UAV, but it has not taken into account that quadrotor may be affected by their own faults in practical applications. The more common faults of quadrotor UAV include actuator failure, sensor, and component failure. The probability of these faults occurring will increase due to harsh working environments, mechanical wear, and other conditions. To ensure flight safety, it is necessary to take into account the faults and external disturbances of quadrotor UAV. The researchers have carried out research on fault-tolerant control of the quadrotors. In Naseri et al.,²⁶ an adaptive obstacle global sliding mode controller based on linear matrix inequalities was designed. By using matrix inequalities, the optimal coefficients of the sliding surface and the adaptive barrier function were determined to handle the problem of actuator failure. In Najafi et al.,²⁷ an adaptive barrier fast terminal sliding mode controller was proposed for quadrotor UAV. By using the adaptive barrier function, the influence of external disturbances, uncertainty terms, and actuator failures were mitigated. In Mai et al.,²⁸ a fault-tolerant control method was proposed based on a nonlinear extended state observer to estimate faults and disturbances, and designed a fast non-singular terminal sliding mode controller to improve the control accuracy. In Wang et al.,²⁹ an approach law for constructing the controller was carried out, which effectively suppressed control buffering and improved the fault-tolerant capability of the UAV by adaptively solving the compensation problems of faults and uncertainties.

In Yu et al.,³⁰ a nonlinear hierarchical control scheme was proposed with a finite-time perturbation observer, effectively suppressing the swing problem of multi-rotor UAV with loads in the case of rotor blade damage. In Li et al.,³¹ a type of neural network interference observer was designed to compensate for external disturbances and actuator failure. Meanwhile, a finite-time controller was designed to control attitude tracking, which improved the fault-tolerant performance of quadrotor UAV. Most of the above-mentioned literatures only consider the partial failure faults of the actuator and ignore the paranoid faults. Therefore, it is very necessary to comprehensively consider the actuator faults, external disturbances and uncertainties, and design a reasonable fault-tolerant control method.

Based on the above analysis, for improving the response speed of quadrotor UAV and having better fault-tolerant capabilities, a non-singular predefined time sliding mode control method is designed for quadcopter UAVs. The main contributions of this article are as follows:

1. To reduce the influence of total disturbances on the quadrotor UAV, a predefined time disturbance observer is designed to estimate the total disturbances. Different from the traditional disturbance observer, the proposed disturbance observer can improve the tracking rate of faults, external disturbances, and uncertainties by adjusting the predefined time parameters. The estimated values are used to compensate the controller of the quadrotor UAV, thereby reducing the impact of disturbances on the quadrotor UAV.
2. To improve the position and attitude tracking performance of the quadrotor, a non-singular predefined time sliding mode controller is designed. By segmenting the traditional predefined time sliding mode surface, a non-singular predefined time sliding mode controller is designed based on the proposed sliding mode surface, and the proposed controller can avoid the singular problem. Through numerical simulation, compared with the traditional predefined time method, the proposed controller has stronger robustness under the influence of faults, external disturbances, and uncertainties.
3. To improve the control performance of control system, the traditional exponential convergence law is improved. A piecewise function is introduced into the exponential convergence term, achieving the function of adjusting according to the segmented conditions during the system approximation process, effectively suppressing the chattering of the system input.

The paper is organized as follows. The dynamical model of quadrotor system is introduced in Section 2. The proposed observer is given in Section 3. The proposed

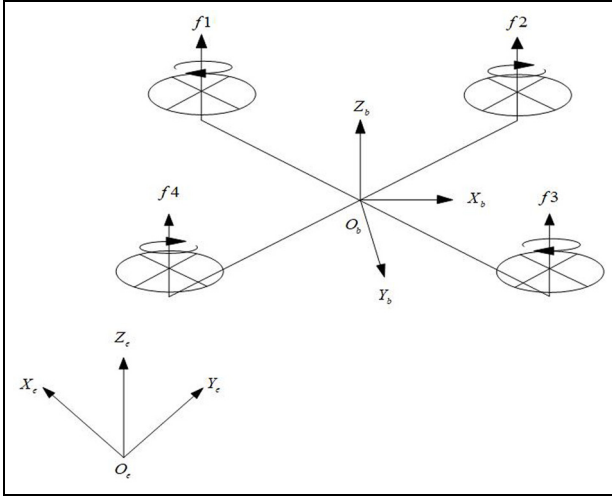


Figure 1. The inertia coordinate system and body coordinate system of quadrotor UAV.

controller with the stability analysis is given in Section 4. In Section 5, the numerical simulation results are given to demonstrate the performance of the proposed control method. Finally, a conclusion is described in Section 6.

Quadrotor dynamic model

The quadrotor UAV adopts an “X” font layout, and its inertia and body coordinate system are shown in Figure 1.

In Figure 1, the propellers 1 and 3 rotate counter-clockwise, and the propellers 2 and 4 rotate clockwise. The coordinate system $O_b X_b Y_b Z_b$ and $O_e X_e Y_e Z_e$ are the body coordinate system and the inertial coordinate system, respectively, expressed as the quadrotor UAV. In order to accurately establish the quadrotor UAV dynamic model with the fault tolerant and design the observer and controller, the following Assumptions and Lemma are adopted.

Assumption 1: The body of quadrotor UAV is rigid and completely symmetrical, and the center of mass of the body coincides with the geometric center of gravity.²

Assumption 2: The attitude Angle of the quadrotor UAV is bounded and satisfied, $-\frac{\pi}{2} < \varnothing < \frac{\pi}{2}$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $-\frac{\pi}{2} < \varphi < \frac{\pi}{2}$.

Lemma 1²⁸: for nonlinear system $\dot{x} = f(t, x)$, if there exists a radially unbounded Lyapunov function that satisfies the following equation:

$$\dot{V}(x) \leq -\frac{\pi}{\eta T_c} (V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}) \quad (1)$$

where, $T_c > 0$ is preset time parameter, $\eta \in (0, 1)$ is adjustable parameter.

Then, the state of the above system $\dot{x} = f(t, x)$ is pre-defined time stable.

Proof:

$$\begin{aligned} \int_{V(0)}^0 \frac{1}{V^{1-\frac{\eta}{2}} + V^{1+\frac{\eta}{2}}} dV &= \int_0^{T(x_0)} -\frac{\pi}{\eta T_c} dt \\ \int_{V(0)}^0 \frac{1}{V^{1-\frac{\eta}{2}}(1+V^\eta)} dV &= -\frac{\pi}{\eta T_c} T(x_0) \\ \int_{V(0)}^0 \frac{2}{\eta(1+V^\eta)} dV^\frac{\eta}{2} &= -\frac{\pi}{\eta T_c} T(x_0) \\ -\frac{2}{\eta} \arctan(V(0)^\frac{\eta}{2}) &= -\frac{\pi}{\eta T_c} T(x_0) \\ T(x_0) &\leq T_c \end{aligned} \quad (2)$$

where, $V(0)$ is the initial value of V , V converges eventually to zero. $T(x_0)$ is convergence time function. From the equation (2), it can be obtain that the convergence time of the system is less than the time parameter T_c .

Considering that the quadrotor UAV is affected by both partial failure and bias failure, the actuator fault model can be expressed as:

$$U_i = p_i u_i + u_{f_i} \quad (3)$$

where, $i \in (1, 2, 3, 4)$, u_i is desired control input, U_i is actual control input, $p_i \in [0, 1]$ is actuator fault factor, u_{f_i} is actuator bias fault. $p_i = 1$ and $u_{f_i} = 0$ indicate the UAV is in normal condition, $p_i \neq 1$ and $u_{f_i} = 0$ indicate the UAV with actuator partial failure only, $p_i = 1$ and $u_{f_i} \neq 0$ indicate the UAV with actuator bias failure only, $p_i \neq 1$ and $u_{f_i} \neq 0$ indicate the UAV with both partial failure and bias failure.²⁴

The dynamics model of the quadrotor UAV with actuator fault can be written as follows:

$$\begin{aligned} \ddot{x} &= \frac{U_1}{m} (\cos\varnothing \cos\varphi \sin\theta + \sin\varnothing \sin\varphi) - \frac{K_1 \dot{x}}{m} + f_1 \\ \ddot{y} &= \frac{U_1}{m} (\cos\varnothing \sin\varphi \sin\theta - \sin\varnothing \cos\varphi) - \frac{K_2 \dot{y}}{m} + f_2 \\ \ddot{z} &= \frac{U_1}{m} (\cos\varnothing \cos\theta) - \frac{K_3 \dot{z}}{m} - g + f_3 \\ \ddot{\varnothing} &= \frac{U_2 - I_r \dot{\varnothing} \Omega_r}{I_{xx}} - \dot{\varphi} \dot{\theta} \frac{I_{zz} - I_{yy}}{I_{xx}} + f_4 \\ \ddot{\theta} &= \frac{U_3 - I_r \dot{\theta} \Omega_r}{I_{yy}} - \dot{\varphi} \dot{\varnothing} \frac{I_{xx} - I_{zz}}{I_{yy}} + f_5 \\ \ddot{\varphi} &= \frac{U_4 - I_r \dot{\varphi} \Omega_r}{I_{zz}} - \dot{\varnothing} \dot{\theta} \frac{I_{yy} - I_{xx}}{I_{zz}} + f_6 \end{aligned} \quad (4)$$

where, g is gravity constant, m is mass of the quadrotor UAV, K_1, K_2, K_3 are air friction resistance coefficient. x, y, z are the position of the quadrotor UAV in inertial coordinate, $\varnothing, \theta, \varphi$ are roll angle, pitch angle, yaw angle of the UAV rotating around the axis X_b, Y_b, Z_b . Ω_r is propeller rotor speed margin, I_r is propeller inertia moment. The UAV is a symmetric rigid body, it can be obtained that the inertia product of the non-parallel axis is 0, and $I = \text{diag}[I_{xx}, I_{yy}, I_{zz}]$. $f_i, i \in (1, 2, 3, 4, 5, 6)$ is bounded external disturbance.

Substitute the equation (3) into the equation (4), and equation (4) can be simplified as:

$$\begin{aligned} \ddot{x} &= ux + d_1 \\ \ddot{y} &= uy + d_2 \\ \ddot{z} &= uz + d_3 \\ \ddot{\varnothing} &= \frac{u_2}{I_{xx}} + d_4 \\ \ddot{\theta} &= \frac{u_3}{I_{yy}} + d_5 \\ \ddot{\varphi} &= \frac{u_4}{I_{zz}} + d_6 \end{aligned} \quad (5)$$

where,

$$\begin{aligned} ux &= \frac{U_1}{m} (\cos\vartheta \cos\varphi \sin\theta + \sin\vartheta \sin\varphi) \\ uy &= \frac{U_1}{m} (\cos\vartheta \sin\varphi \sin\theta - \cos\varphi \sin\vartheta) \\ uz &= \frac{U_1}{m} (\cos\vartheta \cos\theta) - g \end{aligned} \quad (6)$$

$$\begin{aligned} d_1 &= \frac{(p_1-1)u_1 + u_{f1}}{m} (\cos\vartheta \cos\varphi \sin\theta + \sin\vartheta \sin\varphi) - \frac{K_1 \dot{x}}{m} + f_1 \\ d_2 &= \frac{(p_1-1)u_1 + u_{f1}}{m} (\cos\vartheta \sin\varphi \sin\theta - \cos\varphi \sin\vartheta) - \frac{K_2 \dot{y}}{m} + f_2 \\ d_3 &= \frac{(p_1-1)u_1 + u_{f1}}{m} (\cos\vartheta \cos\theta) - \frac{K_3 \dot{z}}{m} - g + f_3 \end{aligned} \quad (7)$$

$$\begin{aligned} d_4 &= \frac{(p_2-1)u_2 + u_{f2} - I_r \dot{\vartheta} \Omega_r}{I_{xx}} - \dot{\varphi} \dot{\theta} \frac{I_{zz} - I_{yy}}{I_{xx}} + f_4 \\ d_5 &= \frac{(p_3-1)u_3 + u_{f3} - I_r \dot{\vartheta} \Omega_r}{I_{yy}} - \dot{\varphi} \dot{\theta} \frac{I_{xx} - I_{zz}}{I_{yy}} + f_5 \\ d_6 &= \frac{(p_4-1)u_4 + u_{f4} - I_r \dot{\varphi} \Omega_r}{I_{zz}} - \dot{\vartheta} \dot{\theta} \frac{I_{yy} - I_{xx}}{I_{zz}} + f_6 \end{aligned} \quad (8)$$

where, $d_i, i \in (1, 2, 3, 4, 5, 6)$ is lumped disturbance consisting of external disturbance, uncertain item, and actuator fault, ux, uy, uz are virtual control item. Since the quadrotor UAV is an underactuated system, it cannot target the six degrees of freedom. Setting the desired position value as x_d, y_d, z_d and desired yaw angle as φ_d , the desired pitch angle θ_d , desired roll angle ϑ_d and u_1 can be derived from the formula (6) as follows:

$$\begin{aligned} \theta_d &= \arctan\left(\frac{\cos\varphi_d ux + \sin\varphi_d uy}{uz + g}\right) \\ \vartheta_d &= \arcsin\left(\frac{m(\sin\varphi_d ux - \cos\varphi_d uy)}{u_1}\right) \\ u_1 &= m\sqrt{ux^2 + uy^2 + (uz + g)^2} \end{aligned} \quad (9)$$

Predefined time disturbance observer design

In order to facilitate the design of the disturbance observer, the equation (5) is rewritten into vector form as follows:

$$\ddot{X} = U + D \quad (10)$$

where, $U = [ux, uy, uz, \frac{u_2}{I_{xx}}, \frac{u_3}{I_{yy}}, \frac{u_4}{I_{zz}}]^T$, $X = [x, y, z, \vartheta, \theta, \varphi]^T$ is the state of the quadrotor UAV, $D = [d_1, d_2, d_3, d_4, d_5, d_6]^T$ is lumped disturbance.

For further improving the control performance of quadrotor UAV, a pre-defined time disturbance observer is designed to estimate the lumped disturbance of quadrotor UAV, and the estimate value of the proposed observer is used as the compensation item for the designed controller. The detailed design steps are as follows.

Designing the auxiliary function as follows:

$$\ddot{Z} = U + \beta E_z \quad (11)$$

where, $Z = [z_1, z_2, z_3, z_4, z_5, z_6]^T$ is the state of the auxiliary function (11), $\beta = \text{diag}[\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6]$ is

positive constant vector, $E_z = [e_{z1}, e_{z2}, e_{z3}, e_{z4}, e_{z5}, e_{z6}]^T = X - Z$ is the error vector of the state of the quadrotor UAV and the state of the auxiliary function.

$$\dot{D} = \beta \dot{E}_z + \ddot{E}_z \quad (12)$$

$$\dot{\hat{E}}_z = \dot{E}_z + \frac{\pi}{2\eta_0 T_0} \left[\alpha_0 \tilde{E}_z^{1-\eta_0} + \tau_0 \tilde{E}_z^{1+\eta_0} \right] + \tilde{E}_z \quad (13)$$

where, $0 \leq \eta_0 < 1$, $\hat{D} = [\hat{d}_1, \hat{d}_2, \hat{d}_3, \hat{d}_4, \hat{d}_5, \hat{d}_6]^T$ is the estimate value of the lumped disturbance D , T_0 is predefined time parameter, $\hat{E}_z = [\hat{e}_{z1}, \hat{e}_{z2}, \hat{e}_{z3}, \hat{e}_{z4}, \hat{e}_{z5}, \hat{e}_{z6}]^T$ is the estimate value of E_z , $\tilde{E}_z = E_z - \hat{E}_z$, $\alpha_0 = \frac{1-\eta_0}{2}$, $\tau_0 = \frac{\eta_0}{2}$.

Under the action of the equations (11) to (13), $\tilde{E}_z$ converge within a predefined time, the process of the proof is as follows:

Subtract $\dot{\hat{E}}_z$ from both sides of equation (13) to obtain the following equation:

$$\dot{\tilde{E}}_z = -\frac{\pi}{2\eta_0 T_0} \left[\alpha_0 \tilde{E}_z^{1-\eta_0} + \tau_0 \tilde{E}_z^{1+\eta_0} \right] - \tilde{E}_z \quad (14)$$

For analyzing the convergence of $\tilde{E}_z$, the Lyapunov function is selected as follows:

$$V_z = \frac{1}{2} \tilde{E}_z^T \tilde{E}_z \quad (15)$$

The time derivative for V_z is:

$$\begin{aligned} \dot{V}_z &= \tilde{E}_z^T \dot{\tilde{E}}_z \\ &= -\frac{\pi}{2\eta_0 T_0} \tilde{E}_z^T \left[\alpha_0 \tilde{E}_z^{1-\eta_0} + \tau_0 \tilde{E}_z^{1+\eta_0} \right] - \tilde{E}_z^T \tilde{E}_z \\ &\leq -\frac{\pi}{\eta_0 T_0} \left(V_z^{1-\frac{\eta_0}{2}} + V_z^{1+\frac{\eta_0}{2}} \right) \end{aligned} \quad (16)$$

According to Lemma 1, the error $\tilde{E}_z$ converges in a predefined time T_0 . Based on equations (11) to (13), the estimate error $\tilde{D} = D - \hat{D}$ satisfies:

$$\begin{aligned} \tilde{D} &= D - \hat{D} \\ &= D - \beta \dot{E}_z - \ddot{X} + \ddot{Z} \\ &= \beta \dot{\tilde{E}}_z \end{aligned} \quad (17)$$

It can be obtained from equations (16) and (17) that the lumped disturbance $\tilde{D}$ satisfies predefined time convergence, when $\tilde{E}_z$ converges within the predefined time.

The controller of quadrotor UAV design

To achieve the control of the quadrotor UAV, the control system of the quadrotor UAV is divided into inner-loop control of attitude and outer-loop control of position. In the position loop and attitude loop, non-singular predefined time controllers are designed, respectively, and the predefined time disturbance

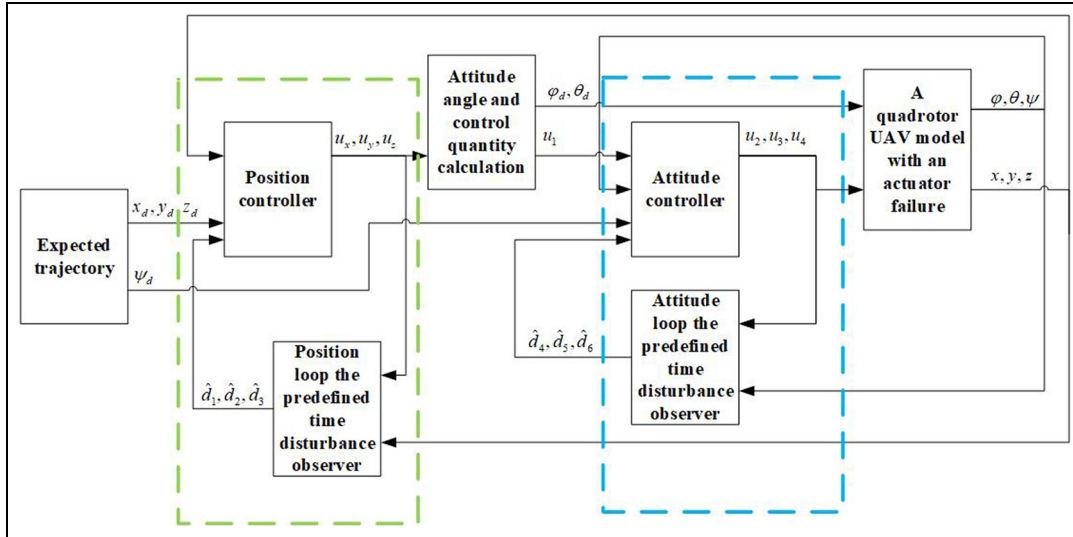


Figure 2. The trajectory tracking control framework of the quadrotor UAV.

observer is used to estimate the disturbances. The trajectory tracking control framework of the quadrotor UAV is shown in Figure 2.

The position controller design

For the quadrotor UAV to quickly track the desired trajectory, a non-singular predefined time sliding mode controller is designed for the position loop. The non-singular predefined time sliding mode surface is designed as follows:

$$s_p = \begin{cases} \dot{e}_p + \frac{\pi}{2\eta_1 T_1} (\alpha_1 e_p^{1-\eta_1} + \tau_1 e_p^{1+\eta_1}), & |e_i| \geq \varepsilon_i \\ \dot{e}_p + \frac{\pi\gamma_1}{2\eta_1 T_1} e_p, & |e_i| < \varepsilon_i \end{cases} \quad (18)$$

where, $i \in (1, 2, 3)$, $0 < \eta_1 < 1$, $s_p = [s_1, s_2, s_3]^T$, $\alpha_1 = \frac{1}{2} \frac{\eta_1}{1-\eta_1}$, $\tau_1 = \frac{1}{2} \frac{\eta_1}{1+\eta_1}$, T_1 is time parameter, ε_i is a very small normal number. To ensure the continuity of the sliding mode surface, the parameter γ_1 satisfies $\gamma_1 = \alpha_1 \varepsilon_i^{-\frac{\eta_1}{2}} + \tau_1 \varepsilon_i^{\frac{\eta_1}{2}}$. $e_p = p - p_d = [e_1, e_2, e_3]^T$ is the position tracking error, $p = [x, y, z]^T$ is the position of the UAV, $p_d = [x_d, y_d, z_d]^T$ is the desired position.

Based on the sliding mode surface (equation (18)), it can be obtained that when $|e_i| \geq \varepsilon_i$, the position tracking error e_p can converge to a small area near zero within a predefined time under the proposed sliding mode surface. When $|e_i| < \varepsilon_i$, the sliding mode surface will be switched, and the position tracking error e_p can asymptotically converge to the origin. The proof is as follows:

When the sliding mode surface $s_i = 0$, and $|e_i| \geq \varepsilon_i$. The time derivative of e_p is:

$$\dot{e}_p = -\frac{\pi}{2\eta_1 T_1} (\alpha_1 e_p^{1-\eta_1} + \tau_1 e_p^{1+\eta_1}) \quad (19)$$

Select the Lyapunov function as $V_1 = \frac{1}{2} e_p^T e_p$, and the time derivative of V_1 is:

$$\begin{aligned} \dot{V}_1 &= e_p^T \dot{e}_p \\ &= -\frac{\pi}{2\eta_1 T_1} e_p^T (\alpha_1 e_p^{1-\eta_1} + \tau_1 e_p^{1+\eta_1}) \\ &= -\frac{\pi}{\eta_1 T_1} (V_1^{1-\frac{\eta_1}{2}} + V_1^{1+\frac{\eta_1}{2}}) \end{aligned} \quad (20)$$

Using the Lemma 1, e_i can converge to ε_i within the time T_1 . When $|e_i| \geq \varepsilon_i$, the error is $e_i(0)$. From the equation (18), the tracking error of the position can be calculated as $e_i = e_i(0) e^{-\frac{\pi\gamma_1}{2\eta_1 T_1} t}$. That is, when $t \rightarrow \infty$, e_i converges exponentially to zero.

Based on the sliding model surface (equation (18)) and the estimation value $\hat{D}$ of the proposed observer, the position controller of the quadrotor UAV is designed as follows:

$$\begin{aligned} \dot{\Psi}_p &= \begin{cases} \frac{\pi}{2\eta_1 T_1} (\alpha_1 (1-\eta_1) e_p^{-\eta_1} + \tau_1 (1+\eta_1) e_p^{\eta_1}) \dot{e}_p, & |e_i| \geq \varepsilon_i \\ \frac{\pi\gamma_1}{2\eta_1 T_1} \dot{e}_p, & |e_i| < \varepsilon_i \end{cases} \\ U_p &= \ddot{p}_d - \dot{\Psi}_p - \hat{D} - \frac{\pi}{2\eta_2 T_2} (\alpha_1 s_p^{1-\eta_2} + \tau_1 s_p^{1+\eta_2}) \\ &\quad - \mu_1 |s_p|^{\delta_1} \text{sign}(s_p) - \mu_2 H(s_p) s_p \end{aligned} \quad (21)$$

where $i \in (1, 2, 3)$, $-\mu_1 |s_p|^{\delta_1} \text{sign}(s_p) - \mu_2 H(s_p) s_p$ is improved exponential convergence law, $H(s_p)$ is piecewise function, and $H(s_p) = |s_p|^{\delta_2 \text{sign}(|s_p|-1)}$. Based on the expression of $H(s_p)$, it can be obtained that when $|s_i| > 1$, $H(s_i) = |s_i|^{\delta_2}$, when $|s_i| < 1$, $H(s_i) = |s_i|^{-\delta_2} > |s_i|^{\delta_2}$. $\hat{D} = [\hat{d}_1, \hat{d}_2, \hat{d}_3]^T$ is the estimation of the lumped perturbation of the position subsystem by the proposed observer. $U_p = [u_x, u_y, u_z]^T$ is the position controller. $\text{sign}(s_p)$ is symbolic function. μ_1 is positive constant,

$0 < \eta_2 < 1$ is adjustable parameter, T_2 is adjustable time parameter. s_p can converge to zero within the time T_2 under the controller U_p .

According to equation (21), it can be obtained that when the exponential term $e_p^{-\eta_1}$ is the factor causing the singular problem, when it is negative value. By segmenting the sliding mode surface, when the exponential term is negative, $e_p \neq 0$. Thus, the singularity problem of the control input is avoided. The stability proof is as follows:

The time derivative of s_p is given as follows:

$$\dot{s}_p = \begin{cases} \ddot{e}_p + \frac{\pi}{2\eta_1 T_1} (\alpha_1 (1 - \eta_1) e_p^{-\eta_1} + \tau_1 (1 + \eta_1) e_p^{\eta_1}) \dot{e}_p, & |e_i| \geq \varepsilon_i \\ \ddot{e}_p + \frac{\pi\gamma_1}{2\eta_1 T_1} \dot{e}_p, & |e_i| < \varepsilon_i \end{cases} \quad (22)$$

Select the Lyapunov function as:

$$V_2 = \frac{1}{2} s_p^T s_p \quad (23)$$

The time derivative of V_2 is given, and substituting equations (21) and (22) into the time derivative of V_2 , as:

$$\begin{aligned} \dot{V}_2 &= s_p^T \dot{s}_p \\ &= s_p^T \left(-\frac{\pi}{2\eta_2 T_2} \left(\alpha_1 s_p^{1-\eta_2} + \tau_1 s_p^{1+\eta_2} \right) - \mu_1 \text{sign}(s_p) \right) \\ &\leq -\frac{\pi}{2\eta_2 T_2} s_p^T \left(\alpha_1 s_p^{1-\eta_2} + \tau_1 s_p^{1+\eta_2} \right) \\ &\leq -\frac{\pi}{\eta_2 T_2} \left(V_2^{1-\frac{\eta_2}{2}} + V_2^{1+\frac{\eta_2}{2}} \right) \end{aligned} \quad (24)$$

Based on the Lemma 1, s_p can converge to zero within the time T_2 . Based on the above analysis, the state of the position loop can achieve convergence to a small area near zero within the time $T_2 + T_1$.

The attitude controller design

For the quadrotor UAV to maintain the attitude stability, a non-singular predefined time sliding mode controller is designed for the attitude loop. The non-singular predefined time sliding mode surface is designed as follows:

$$s_a = \begin{cases} \dot{e}_a + \frac{\pi}{2\eta_3 T_3} \left(\alpha_2 e_a^{1-\eta_3} + \tau_2 e_a^{1+\eta_3} \right), & |e_j| \geq \varepsilon_j \\ \dot{e}_a + \frac{\pi\gamma_2}{2\eta_3 T_3} e_a, & |e_j| < \varepsilon_j \end{cases} \quad (25)$$

where, $j \in (4, 5, 6)$, $0 < \eta_3 < 1$, $s_a = [s_4, s_5, s_6]^T$, $\alpha_2 = \frac{1}{2} e^{-\frac{\eta_3}{2}}$, $\tau_2 = \frac{1}{2} e^{\frac{\eta_3}{2}}$, T_3 is time parameter, ε_j is a small positive constant. To ensure the continuity of the sliding mode surface, the parameter γ_2 satisfies $\gamma_2 = \alpha_2 \varepsilon_j^{-\frac{\eta_3}{2}} + \tau_1 \varepsilon_j^{\frac{\eta_3}{2}}$. $a = [\emptyset, \theta, \varphi]^T$ is the attitude angle

of the quadrotor UAV, $a_d = [\emptyset_d, \theta_d, \varphi_d]^T$ is the desired attitude, $e_a = a - a_d = [e_4, e_5, e_6]^T$ is the attitude tracking error.

When $|e_j| \geq \varepsilon_j$, the attitude tracking error e_j can converge to a small area near zero within a predefined time under the proposed sliding mode surface. When $|e_j| < \varepsilon_j$, the sliding mode surface is switched, and the attitude tracking error asymptotic converges to the origin. The proof process is as follows:

When $s_j = 0$ and $|e_j| \geq \varepsilon_j$, the follow equation can be expressed as follows:

$$\dot{e}_a = -\frac{\pi}{2\eta_3 T_3} \left(\alpha_2 e_a^{1-\eta_3} + \tau_2 e_a^{1+\eta_3} \right) \quad (26)$$

Select the Lyapunov function as $V_3 = \frac{1}{2} e_a^T e_a$, and derivative V_3

$$\begin{aligned} \dot{V}_3 &= e_a^T \dot{e}_a \\ &= -\frac{\pi}{2\eta_3 T_3} e_a^T \left(\alpha_2 e_a^{1-\eta_3} + \tau_2 e_a^{1+\eta_3} \right) \\ &= -\frac{\pi}{\eta_3 T_3} \left(V_3^{1-\frac{\eta_3}{2}} + V_3^{1+\frac{\eta_3}{2}} \right) \end{aligned} \quad (27)$$

Based on the Lemma 1, e_j can converge to ε_j . When $|e_j| < \varepsilon_j$, the error is $e_j(0)$, and based on equation (25), the tracking error of attitude can be calculated as $e_j = e_j(0) e^{-\frac{\pi\gamma_2}{2\eta_3 T_3} t}$. That is, when $t \rightarrow \infty$, e_j converges exponentially to zero:

$$\begin{aligned} \dot{\Psi}_a &= \begin{cases} \frac{\pi}{2\eta_3 T_3} (\alpha_2 (1 - \eta_3) e_a^{-\eta_3} + \tau_2 (1 + \eta_3) e_a^{\eta_3}) \dot{e}_a, & |e_j| \geq \varepsilon_j \\ \frac{\pi\gamma_2}{2\eta_3 T_3} \dot{e}_a, & |e_j| < \varepsilon_j \end{cases} \\ U_a &= \ddot{a}_d - \dot{\Psi}_a - \hat{D}_2 - \frac{\pi}{2\eta_4 T_4} (\alpha_2 s_a^{1-\eta_4} + \tau_2 s_a^{1+\eta_4}) \\ &\quad - \mu_3 |s_a|^{\delta_3} \text{sign}(s_a) - \mu_4 H(s_a) s_a \end{aligned} \quad (28)$$

where $j \in (4, 5, 6)$, $-\mu_3 |s_a|^{\delta_3} \text{sign}(s_a) - \mu_4 H(s_a) s_a$ is improved exponential convergence law, $H(s_a)$ is piecewise function, and $H(s_a) = |s_a|^{\delta_4 \text{sign}(|s_a|-1)}$. Based on the expression of $H(s_a)$, it can be obtained that when $|s_j| > 1$, $H(s_j) = |s_j|^{\delta_4}$, when $|s_j| < 1$, $H(s_j) = |s_j|^{-\delta_4} > |s_j|^{\delta_4}$. $\hat{D}_2 = [\hat{d}_4, \hat{d}_5, \hat{d}_6]^T$ is the estimation of the lumped disturbance of the attitude loop by the disturbance observer. $\text{sign}(s_a)$ is symbolic function. μ_2 is positive constant, $0 < \eta_4 < 1$ is adjustable parameter, T_4 is adjustable time parameter. s_a can converge to zero within the time T_4 under the controller U_a .

According to equation (28), it can be obtained that when the exponential term $e_a^{-\eta_3}$ is the factor causing the singular problem, when it is negative value. By segmenting the sliding mode surface, when the exponential term $e_a^{-\eta_3}$ is negative, $e_a \neq 0$. Thus, the singularity problem of the attitude controller $U_a = \left[\frac{u_2}{I_{xx}}, \frac{u_3}{I_{yy}}, \frac{u_4}{I_{zz}} \right]^T$ is avoided.

The stability proof is as follows:

The time derivative of s_a is given as follows:

$$\dot{s}_a = \begin{cases} \ddot{e}_a + \frac{\pi}{2\eta_3 T_3} (\alpha_2(1 - \eta_3)e_a^{-\eta_3} + \tau_2(1 + \eta_3)e_a^{\eta_3})\dot{e}_a, & |e_j| \geq \varepsilon_j \\ \ddot{e}_a + \frac{\pi\gamma_2}{2\eta_3 T_3} \dot{e}_a, & |e_j| < \varepsilon_j \end{cases} \quad (29)$$

Select the Lyapunov function as:

$$V_4 = \frac{1}{2} s_a^T s_a \quad (30)$$

The time derivative of equation (30) is given, and substituting equations (28) and (29) into the time derivative of V_4 , as:

$$\begin{aligned} \dot{V}_4 &= s_a^T \dot{s}_a \\ &= s_a^T \left(-\frac{\pi}{2\eta_4 T_4} \left(\alpha_2 s_a^{1-\eta_4} + \tau_2 s_a^{1+\eta_4} \right) - \mu_2 \text{sign}(s_a) \right) \\ &\leq -\frac{\pi}{2\eta_4 T_4} s_a^T \left(\alpha_2 s_a^{1-\eta_4} + \tau_2 s_a^{1+\eta_4} \right) \\ &\leq -\frac{\pi}{\eta_4 T_4} \left((V_4)^{1-\frac{\eta_4}{2}} + V_4^{1+\frac{\eta_4}{2}} \right) \end{aligned} \quad (31)$$

Based on the Lemma 1, s_a can converge to zero within the time T_4 . Based on the above analysis, the state of the attitude loop can achieve convergence to a small area near zero within the time $T_4 + T_3$.

Numerical simulation

In order to verify the effectiveness of the proposed control scheme proposed in the paper, The numerical simulations are carried out for quadrotor UAV with actuator failures by using the MatLab/Simulink. In the numerical simulations, the main parameters of the quadrotor UAV are set as follows: the acceleration of gravity $g = 9.8m \cdot s^{-2}$, total mass $m = 2kg$, the air resistance coefficient is set as $k_i = 0.05N \cdot s \cdot m^{-1}$, $i \in (1, 2, 3)$, the rotational inertias are set as $I_{xx} = 2.5kg \cdot m^2$, $I_{yy} = I_{zz} = 1.25kg \cdot m^2$, the inertial constant of the rotor is $I_r = 1kg \cdot m^2$, rotor speed margin $\Omega_r = 1rad$. Considering that the quadrotor UAV is suffering from the external disturbances, the external disturbances of the position loop are set as $f_1 = f_2 = f_3 = 0.1\sin(\frac{2\pi t}{3})$, and the external disturbances of the attitude loop are set as $f_4 = 0.05\sin(t) + 0.1$, $f_5 = f_6 = 0.1\sin(t) + 0.3$. The initial values of the position are set as $x_0 = y_0 = z_0 = 0.1$, and the initial values of the attitude are set as $\theta_0 = \theta_0 = \varphi_0 = \frac{\pi}{6}$. The desired values of the position and attitude are $x_d = y_d = z_d = \cos(t) - 0.7$ and $\theta_d = \cos(t) - 0.2$, respectively. The main parameters of the proposed observer are $\beta = [0.1, 0.1, 0.1, 0.1, 0.1, 0.1]^T$, $i \in (1, 2, 3)$, $\eta_0 = 0.3$, $T_0 = 1s$. The main parameters of the proposed controller are $\varepsilon_j = \varepsilon_i = 0.001$, $j \in (4, 5, 6)$, $T_1 = T_2 = 3s$, $T_3 = T_4 = 1s$, $\eta_1 = \eta_2 = 0.3$, $\eta_3 = \eta_4 = 0.3$, $\mu_1 = 0.01$, $\delta_1 = 0.3$, $\mu_2 = 0.1$, $\delta_2 = 0.3$, $\mu_3 = 2$, $\delta_3 = 0.5$, $\mu_4 = 60$, $\delta_4 = 0.3$.

Considering that the quadrotor UAV is affected by actuator failure, the actuator failure of the quadrotor UAV is set at $t \in (5.5, 6.5)$, and partial failure parameters are $p_1 = 1$, $p_2 = p_3 = p_4 = 0.7$. The bias parameters are $uf_i = 0.6$, $i \in (1, 2, 3, 4)$.

In order to verify the superiority of the control method proposed in this paper, a comparative simulation experiment is conducted under the same conditions of the quadrotor system model, disturbances, faults, and system initial state settings. The control method (NPTSM2) proposed in this paper is compared with the following three methods: the predefined time sliding mode control (PTSM) in Wu et al.,³² the non-singular terminal sliding mode control (NTSMC) in Hou et al.,³³ and the NPTSM1 control method based on the traditional exponential convergence law.

Figures 3 and 4 show the estimation curves of the proposed observer for the lumped perturbation of position and attitude, respectively. It can be observed from Figures 3 and 4 that the proposed observer can estimate the lumped disturbances within 1 s, and the convergence time is within the predefined time T_0 , and the validity of the proposed disturbance observer is verified. Similarly, it also shows that during the actuator failure period, the proposed observer can estimate the sudden lumped perturbation well, indicating that the designed observer has outstanding robustness.

Figures 5 to 8 show the trajectory curves and error curves of the position and orientation tracking of the quadrotor unmanned aerial vehicle under different control algorithms. From the figures, it can be seen that under the NPTSM2 and NPTSM1 methods, the convergence time of the tracking errors for x, y, z channels are 0.8, 1.2, and 1.3 s, respectively, and the steady-state error reaches the order of 10^{-4} , and the convergence time of the tracking errors for $\theta, \emptyset, \varphi$ channels are 0.5, 0.5, and 0.3 s, respectively, and the steady-state error reaches the order of 10^{-4} , with an overshoot of 0.005 in $\theta, \emptyset$ channels. Under the PTSM control method, the convergence time of the tracking errors for x, y, z channels are 0.6, 1.2, and 1.3 s, respectively, and the steady-state error reaches the order of 10^{-4} . The convergence time of the tracking errors for $\theta, \emptyset, \varphi$ channels are 0.8, 0.7, and 0.4 s, respectively, and the steady-state error reaches the order of 10^{-4} , with an overshoot of 0.02 in $\theta, \emptyset$ channels. It can also be seen that during the occurrence of actuator failures, the steady-state error of $\theta, \emptyset$ channels shows obvious fluctuations. Under the NTSMC control method, the convergence time of the tracking errors for x, y, z channels are 2.5, 3.5, and 2.5 s, respectively, and the steady-state error reaches the order of 10^{-2} . The convergence time of the tracking errors for $\theta, \emptyset, \varphi$ channels are 2.4, 2.3, and 2 s, respectively, and the steady-state error reaches the order of 10^{-2} , with an overshoot of 0.15 for x, z channels. It can also be observed that under the NTSMC method, during the period of actuator failure, the steady-state errors of each states of the quadrotor UAV exhibit significant fluctuations. Based on the above simulation

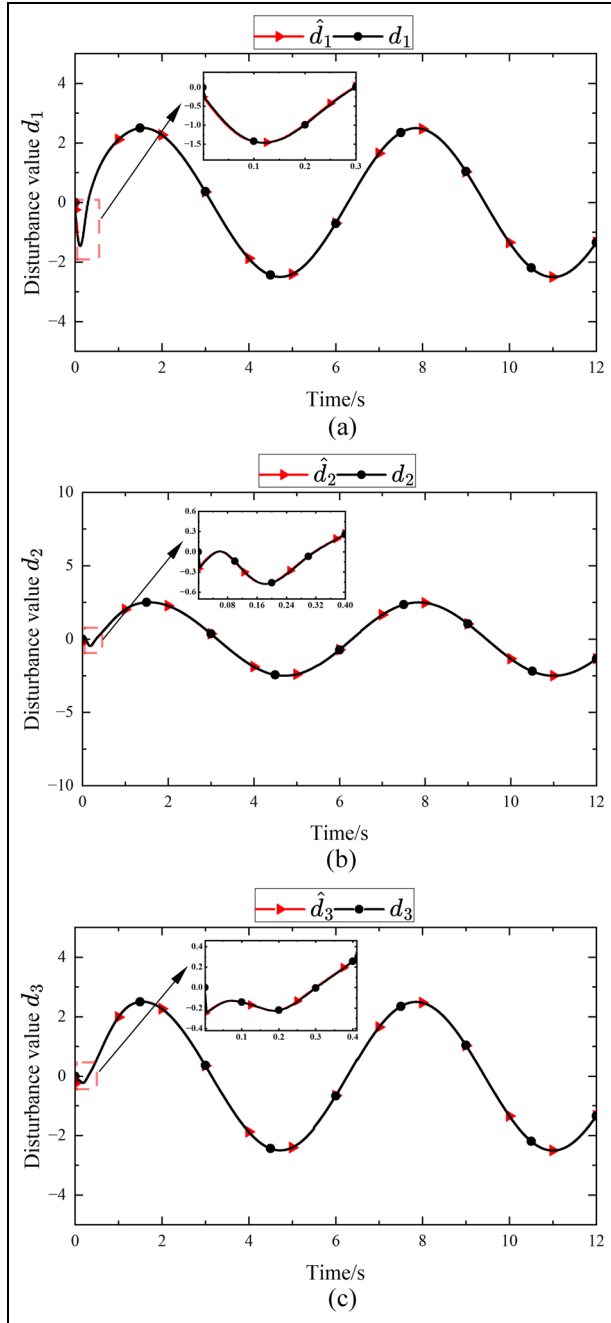


Figure 3. The lump disturbance estimation curve of position. (a) Estimation curve of disturbance d_1 . (b) Estimation curve of disturbance d_2 . (c) Estimation curve of disturbance d_3 .

results analysis, the NPTSM2 method, the NPTSM1 method, and the PTSM method can all enable the quadrotor UAV to complete the trajectory tracking within the predefined time. Compared with the PTSM method, under the NPTSM2 method, the convergence time of the attitude is shorter and it shows stronger robustness to faults. Compared with the NTSMC method, under the NPTSM2 method, the convergence times of the position and attitude are shorter, the overshoots are smaller, the steady-state error accuracy is higher, and better robustness to faults.

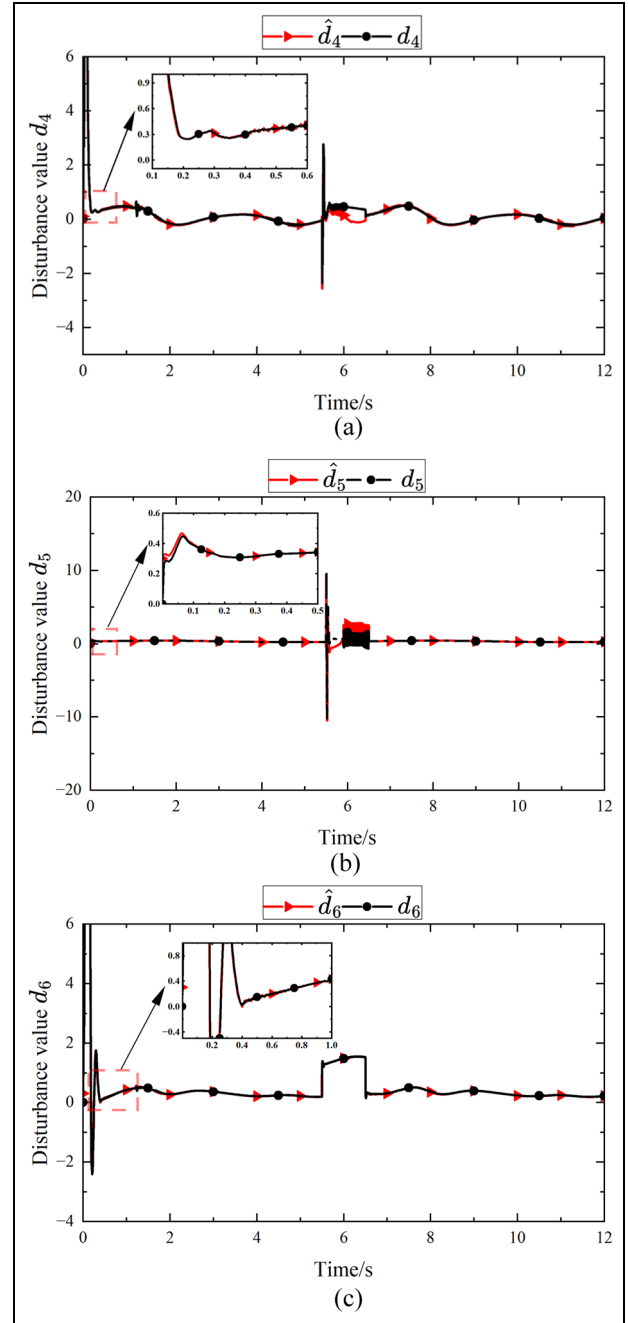


Figure 4. The lump disturbance estimation curve of attitude. (a) Estimation curve of disturbance d_4 . (b) Estimation curve of disturbance d_5 . (c) Estimation curve of disturbance d_6 .

Figure 9 shows the control inputs of the four control method. From Figure 9, it can be seen that compared with NPTSM1, the chattering amplitude of control input of the attitude under the NPTSM2 method proposed in this paper has significantly decreased, which proves that the improved exponential convergence law in this paper has a better damping suppression effect than the traditional exponential convergence law. Compared with NPTSM2 method, the PTSM and NTSMC methods have longer and higher chattering amplitude in the control input of the roll and pitch

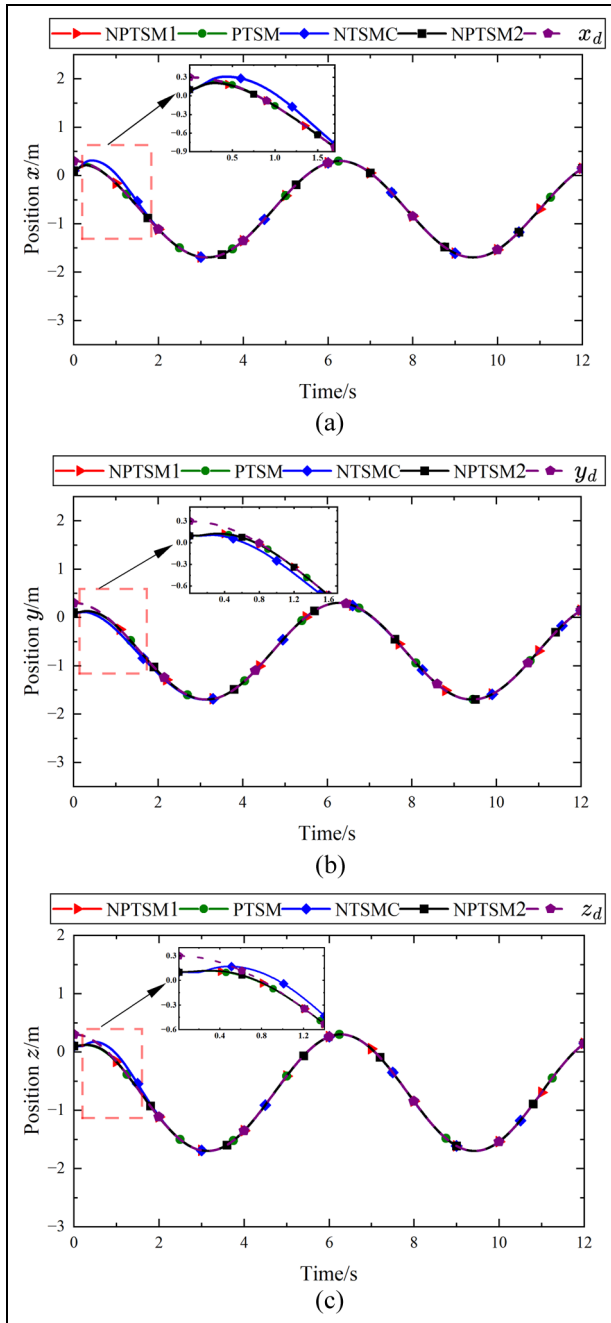


Figure 5. Tracking curve of the position. (a) Tracking curve of x . (b) Tracking curve of y . (c) Tracking curve of z .

channels. In summary, based on the simulation results, the NPTSM2 method designed in this paper can complete the tracking of the desired value within the predefined time, and has a faster convergence rate and effective damping of chattering. At the same time, the proposed control method has better robustness when the quadrotor UAV is suffering from actuator failure, external disturbances, and uncertain terms.

Conclusion

This paper studies the problem that quadrotor UAV vulnerable to disturbances and actuator failures, and a

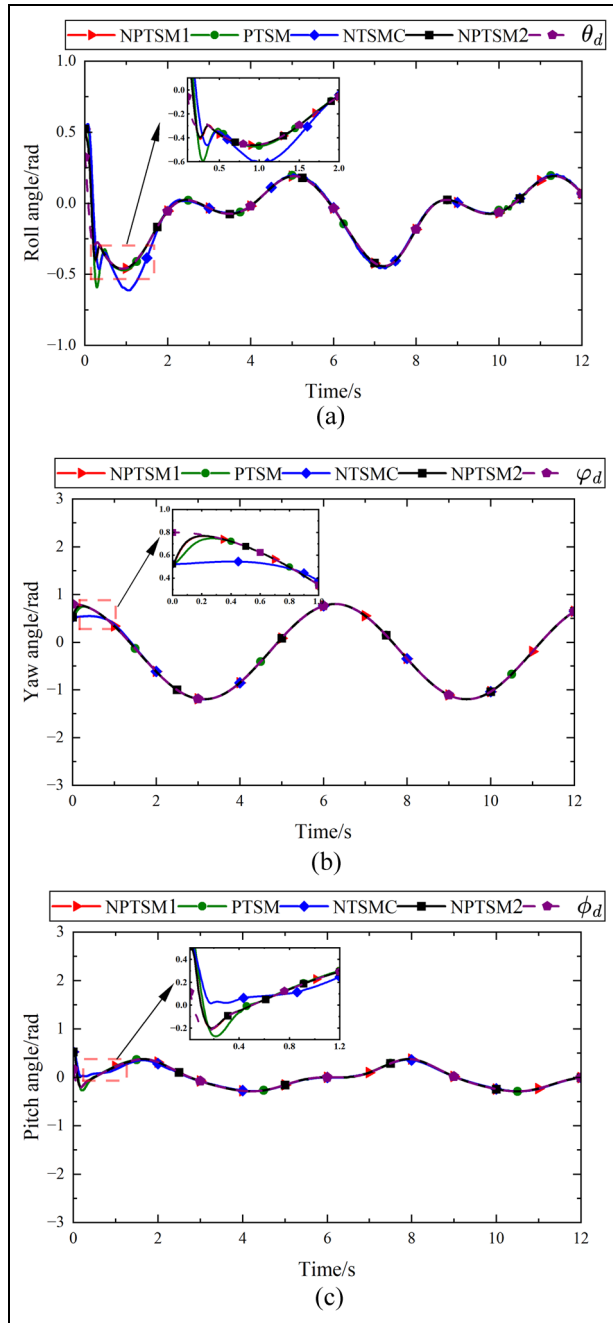


Figure 6. Tracking curve of attitude. (a) Tracking curve of roll angle. (b) Tracking curve of yaw angle. (c) Tracking curve of pitch angle.

non-singular predefined time sliding mode control method is designed. Through the analysis of the actuator failure model and the quadrotor UAV model, actuator failures and external disturbances are regarded as lumped disturbances, thereby, the quadcopter UAV with failure model is established. In order to reduce the influence of lumped perturbation on quadcopter UAV, a predefined time disturbance observer is designed based on the predefined time stability theory. To address the singularity problem of the traditional predefined time sliding mode control, a non-singular predefined time sliding mode surface is designed by segmenting the

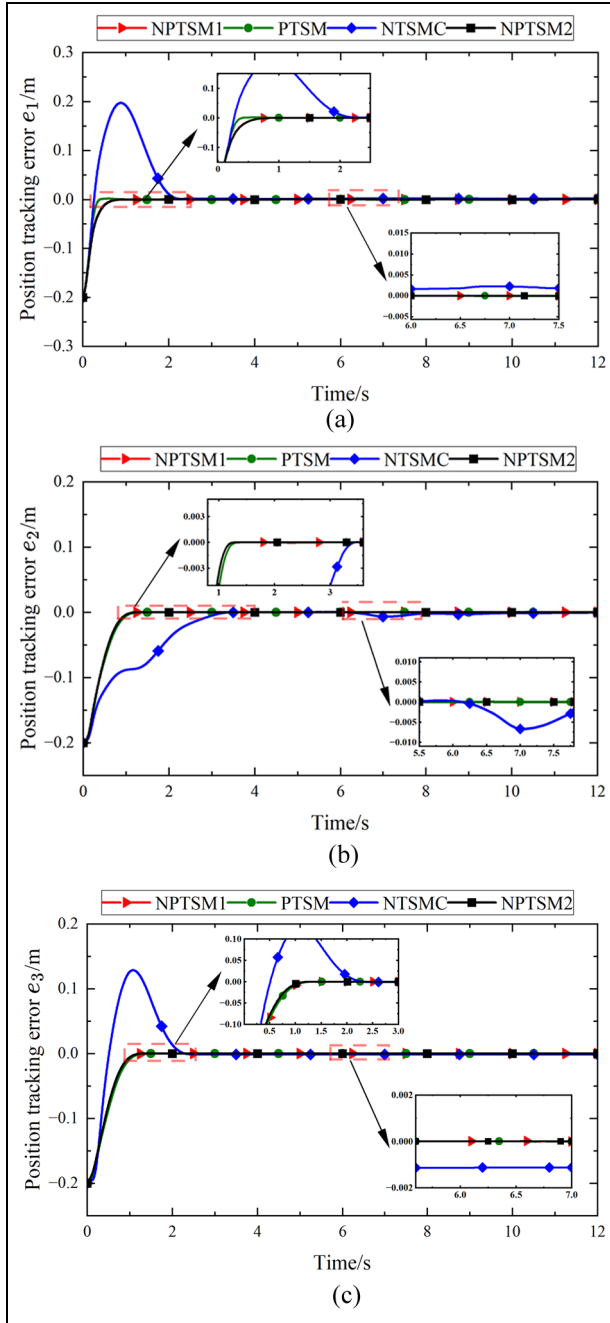


Figure 7. Position tracking error curve. (a) Position tracking error e_1 . (b) Position tracking error e_2 . (c) Position tracking error e_3 .

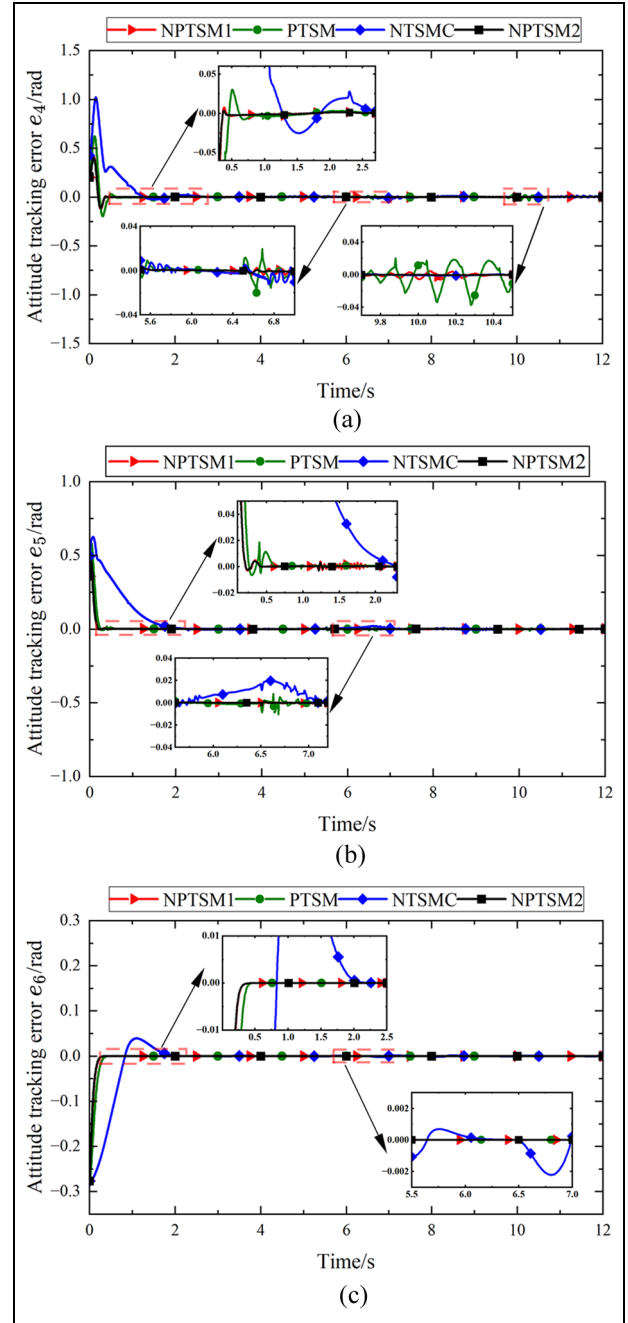


Figure 8. Attitude tracking error curve. (a) Attitude tracking error e_4 . (b) Attitude tracking error e_5 . (c) Attitude tracking error e_6 .

sliding mode surface. At the same time, to solve the problem of control input chattering, the traditional exponential convergence law is improved. The proposed sliding mode surface and the convergence law enable the control input to avoid the singularity problem and suppress the jitter. The stability and convergence of the designed controller and observer are proved by Lyapunov stability theory and Lemma 1. The simulation results show that the control algorithm designed in

this paper has a faster convergence rate, robustness and chattering suppression effect. Hence, the proposed control method can better handle the actuator faults and disturbances of quadrotor UAV, and improve the convergence speed and fault-tolerant ability of the quadrotor UAV. However, there are still some potential problems that have not been considered, such as output delay during flight. In the future, these issues will be committed to working out reasonable solutions.

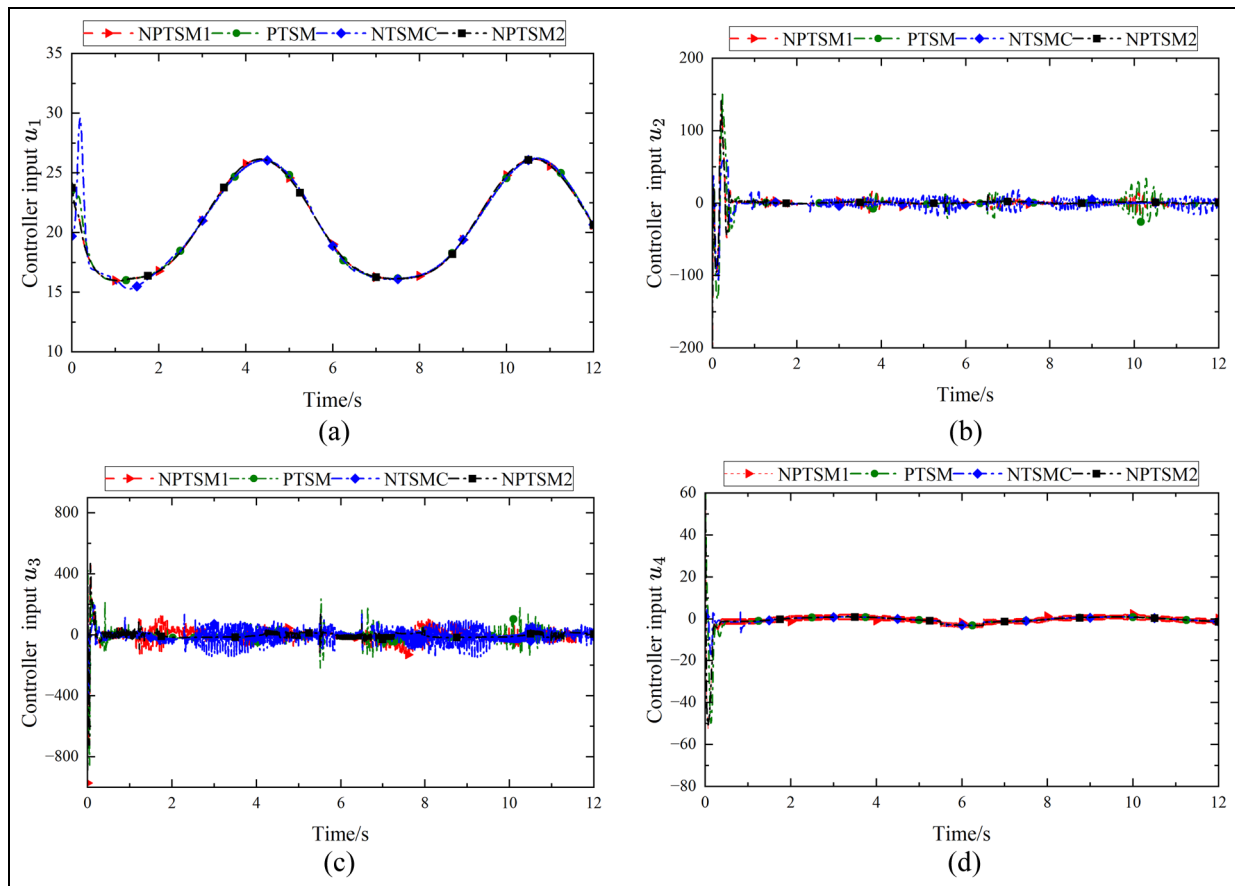


Figure 9. Controller input. (a) Controller input u_1 . (b) Controller input u_2 . (c) Controller input u_3 . (d) Controller input u_4 .

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Ethical considerations

This study is a quadrotor UAV modeling control scheme analysis and does not involve human participants, animal experiments, or personal identifiable data.

Consent to participate

No human/animal data were collected or analyzed in this research.

Consent to publication

This is to confirm that the manuscript titled “*Non-singular Predefined Time Sliding Mode Control for Quadrotor UAV with Actuator Fault*,” we are submitting has never been published before. Also, it is not currently being considered for publication in any other journal. All the listed authors have carefully reviewed and approved this manuscript. We all agree to submit it to *Measurement and Control* for publication and are aware of and accept the journal’s publication policies.

Author contributions

All authors contributed to the study conception and design, data collection and analysis were performed by Jingxin Dou, Dongwu Xie, Yingliang Wu, and Ti Zhang. The first draft of the manuscript was written by Jingxin Dou and all authors

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The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

Within the scope of this study, all utilized data and relevant materials are available upon request from the authors.

Trial registration number/date

This study did not involve a clinical trial requiring registration.

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